Amrit Gopinath

B.E. Computer Science & Engineering · AI/ML Research

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Education

Sri Sivasubramaniya Nadar College of Engineering, Chennai 2024–2028 (expected)
B.E. Computer Science and Engineering; Honors track in AI/ML.
CGPA: 9.383/10.0 | **Department rank: 1st** (through Semester 4).

Research Interests

Multilingual language models, expert routing, and representation analysis; causal interpretability and model introspection, with interests in vision–language models and AI safety.

Selected Research

A Declarative–Procedural Perspective on Expert Routing in Bilingual Mixture-of-Experts Language Models 2026
First author | ARR 2026 submission | arXiv:2608.15102 | Code

- Investigated whether English–German MoE routing reflects language identity and lexical information or higher-level grammatical and syntactic structure.
- Designed controlled bilingual corpora, curriculum and no-curriculum training, and held-out lexical, grammatical, and syntactic probes; analyzed routing with mutual and conditional mutual information, entropy, Jensen–Shannon distance, frequency controls, and targeted ablations.
- Found stronger aggregate specialization without a curriculum (peak MI **0.2599** versus **0.1148**), but second-seed replication exposed seed-dependent language dominance; curriculum training produced a more language-balanced routing profile across the two seeds.
- Led the research question, experimental design, analysis pipeline, and manuscript preparation.

Having a State Is Not Knowing It 2026
Co-author | Apart Research Digital Minds Research Sprint | Report | Project

- Investigated model introspection through a falsifiable hierarchy separating causal state presence, verbal report, counterfactual intervention prediction, and quantitative self-modeling.
- Replicated concept injection in Llama-3.2-3B-Instruct and built a controlled Qwen decision-state benchmark using measured logit-margin shifts as ground truth. Naive self-report achieved **0.243 macro-F1**; trained native reporting reached **0.522**, while a matched external residual MLP achieved **1.000 F1** on this controlled benchmark.
- Predicted intervention direction at **0.695 F1 on 96 held-out base trials**; performance fell to **0.564** on unseen direction mixtures. Negative response-curve R^2 separated conditional qualitative sensitivity from quantitative self-modeling.

Research Experience

NLP Research Intern, National Institute of Technology Tiruchirappalli 2026
Faculty mentorship | First-author manuscript submitted to ICON 2026 | Paper | Code

- Studied Classical Tamil verse–commentary representations using a corpus of **1,262 pairs**; developed recurrent and Transformer encoders, Siamese matching, and encoder–decoder and decoder-only generation pipelines.
- Evaluated learned representations and generation against lexical, random-data, and constant-output controls; diagnosed inflated canonical correlation and overfitting rather than treating them as evidence of semantic alignment.
- Found authentic word-order preference in **107/112 minimal-pair comparisons (95.5%)**, while held-out and out-of-domain evaluations exposed failures to generate verse-specific commentary.

Publications and Benchmark Research

Exploring Swin Transformers, Luminance Input, and Hybrid Architectures for Synthetic Image Detection 2026

MediaEval 2026 Working Notes | First author | Paper | Code

- Compared Swin, global and MIL-style aggregation, luminance-only inputs, and CNN/ResNet–GRU–ViT hybrids for real-versus-synthetic image classification.
- Swin Base achieved **0.6997 synthetic-class F1** and **0.7471 AUC** in the **open validation setting** with external SID_Set training data; analyzed precision–recall differences between open and constrained runs.

FinMMEval 2026: TextSentinels 2026

CLEF 2026 Working Notes (CEUR-WS, Vol. 4283) | 3 published task papers | Co-author

- Contributed to multilingual financial MCQA, grounded short-answer QA, and decision-making systems spanning question-type model routing, FAISS retrieval, BM25 over SEC filings and tables, and sentiment-aware trading.
- Financial QA improved from approximately **0.04 to 0.28 ROUGE-1 F1 on local validation**; the final benchmark submission achieved approximately **0.18**. Historical backtests across six assets yielded a **mean per-asset annualized Sharpe ratio of 2.03**.

Published task papers (titles below share the prefix "TextSentinels @ FinMMEval 2026"):

Task 1: A Multilingual Routed Retrieval-Augmented System for Financial Multiple Choice Questions.
pp. 2635–2643.

Task 2: Chronological Stabilization of a BM25-Grounded Multilingual Financial Question Answering Pipeline.
pp. 2644–2651.

Task 3: Multi-Asset Financial Decision Making Using Machine Learning and Financial Transformers.
pp. 2624–2634.

Technical Skills

ML / NLP	PyTorch, Hugging Face Transformers, scikit-learn, FAISS, Stanza.
Research methods	Mutual and conditional mutual information, routing entropy, Jensen–Shannon distance, permutation tests, concept injection, external probes, counterfactual interventions; CCA with matched null controls.
Programming / tools	Python, C, Java, TypeScript, Git, Linux, LaTeX.

Academic Distinctions and Service

- **Academic standing:** department rank 1st through Semester 4; CGPA 9.383/10.0.
- **Yearly GPA awards: 2nd position for second-year GPA**; 3rd position for first-year GPA.
- **Reviewing assistance, CLEF 2026 Working Notes:** assessed methodological clarity, experimental validity, presentation, and reproducibility.